AIMS5701: Fundamentals in AI

Homework 2: Bayes Nets I

Deadline: October 23, 2026

Problems

All numerical probabilities below are hypothetical teaching values. All variables are binary, with value 1 indicating that the named event occurs. Omitted probabilities are complements. Symbols are defined locally in each problem.

Problem 1: Probability Warm-up (15 marks)

This problem reviews basic rules that underlie Bayes nets.

(a) Show that for random variables X, Y, Z, the chain rule can be written as

$$P(X, Y, Z) = P(X)\,P(Y \mid X)\,P(Z \mid X, Y).$$

Then list two other valid decompositions by changing the variable ordering. (5 marks)

(b) **Diagnostic inference.** Let F mean that a device's battery is faulty, and W mean that the device displays a battery warning. Suppose

$$P(F) = 0.002, \qquad P(W \mid F) = 0.90, \qquad P(W \mid \neg F) = 0.04.$$

Compute $P(F \mid W)$ exactly and report it to at least three significant digits. Briefly interpret why a warning does not establish that the battery is faulty. (10 marks)

Problem 2: Bayes Net Semantics & Joint Factorization (20 marks)

Consider a simplified model of air-conditioning use in a dormitory during a fixed observation period:

$$H \to C \to E, \qquad C \to N,$$

where H = high outdoor temperature, C = air conditioner on, E = high electricity consumption, and N = audible air-conditioner noise. Assume the dependencies are exactly those specified by this graph. The CPTs are:

$$\begin{aligned}
&P(H{=}1) = 0.40, \qquad P(H{=}0) = 0.60;\\
&P(C{=}1 \mid H{=}1) = 0.85, \qquad P(C{=}1 \mid H{=}0) = 0.15;\\
&P(E{=}1 \mid C{=}1) = 0.80, \qquad P(E{=}1 \mid C{=}0) = 0.25;\\
&P(N{=}1 \mid C{=}1) = 0.70, \qquad P(N{=}1 \mid C{=}0) = 0.05.
\end{aligned}$$

(a) Write the full joint distribution $P(H, C, E, N)$ as a product of the local conditional distributions implied by the graph. (5 marks)

(b) Compute $P(C{=}1)$ and $P(E{=}1)$ by appropriately marginalizing the joint. Show your intermediate steps, including the sums. (8 marks)

(c) Compute $P(C{=}1 \mid E{=}1)$ using Bayes' rule and your previous results. (7 marks)

Problem 3: Parameter Counting & Structure Choice (15 marks)

Assume all variables in a Bayes net are binary.

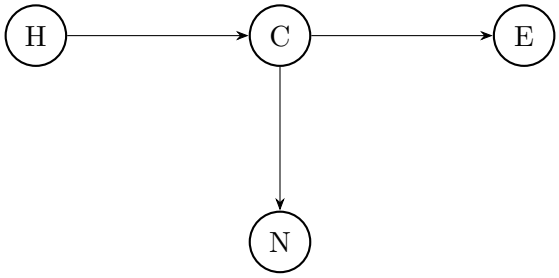
(a) For the graph in Problem 2, how many free parameters are required to specify all CPTs? Show your counting clearly for each node. (7 marks)

(b) Consider the alternative two-node structure $E \to C$, with H, N removed. Interpreted causally, this reverses the direction from air-conditioner use to electricity consumption. How many parameters does this alternative require? Which of the two structures (Problem 2 vs. this alternative) is likely easier to elicit from experts, and why? (8 marks)

Problem 4: D-Separation & Conditional Independence (20 marks)

Use d-separation to answer the following questions. State whether each independence is *guaranteed by the graph*, without relying on special parameter values, and justify your answer by naming the active or blocked paths. Graph A is the dormitory model from Problem 2.

Graph A

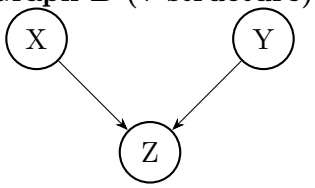


(a) In Graph A, is $H \perp\!\!\!\perp E$ guaranteed? Is $H \perp\!\!\!\perp E \mid C$ guaranteed? Explain using the causal chain and blocking by an observed middle node. (8 marks)

(b) In Graph A, is $N \perp\!\!\!\perp H \mid C$ guaranteed? Is $N \perp\!\!\!\perp E \mid C$ guaranteed? Explain how conditioning on parents renders a node independent of its non-descendants. (6 marks)

Now consider a simplified refrigerator model: X = door left open, Y = faulty thermostat, and Z = high refrigerator temperature. The two causes X, Y are modeled as independent roots in Graph B:

Graph B (v-structure)



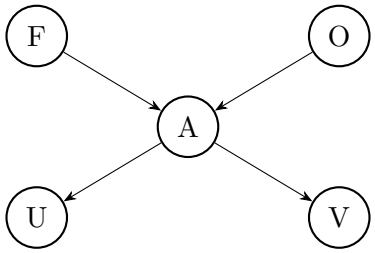
(c) In Graph B, is $X \perp\!\!\!\perp Y$ guaranteed? Is $X \perp\!\!\!\perp Y \mid Z$ guaranteed? Explain the "explaining away" phenomenon using the refrigerator scenario. (6 marks)

Problem 5: Home Network Fault Reasoning (15 marks)

Consider the following simplified home network model:

$$F \to A \leftarrow O, \qquad A \to U, \qquad A \to V,$$

where F = router fault, O = internet service provider outage, A = home internet access unavailable, U = computer connection warning, and V = phone connection warning. All events refer to the same observation period. The roots are independent; the two device warnings are conditionally independent given A.



Use the following CPTs:

$$\begin{aligned}
&P(F{=}1) = 0.01, \qquad P(O{=}1) = 0.02;\\
&P(A{=}1 \mid F{=}1, O{=}1) = 0.98, \quad P(A{=}1 \mid F{=}1, O{=}0) = 0.95;\\
&P(A{=}1 \mid F{=}0, O{=}1) = 0.90, \quad P(A{=}1 \mid F{=}0, O{=}0) = 0.005;\\
&P(U{=}1 \mid A{=}1) = 0.92, \qquad P(U{=}1 \mid A{=}0) = 0.03;\\
&P(V{=}1 \mid A{=}1) = 0.85, \qquad P(V{=}1 \mid A{=}0) = 0.02.
\end{aligned}$$

(a) Compute $P(A{=}1)$ by marginalizing over F, O. Show your sums. (6 marks)

(b) Compute $P(F{=}1 \mid U{=}1, V{=}1)$ exactly by expanding the numerator and denominator over A, O as needed. You may leave intermediate arithmetic as sums and products, but provide a final numerical value. (9 marks)

Problem 6: Guaranteed Independences from Structure (15 marks)

(a) State and explain the general rule: "In a Bayes net, a node is conditionally independent of its non-descendants given its parents." Illustrate it on Graph A from Problem 4 by listing two concrete independences entailed by the structure. (6 marks)

(b) For the following graph, list *all pairwise conditional independences* guaranteed solely by d-separation, without assuming special parameter values:

$$X \to Y, \qquad X \to Z, \qquad Y \to W.$$

Write each answer as $U \perp\!\!\!\perp V \mid S$, where U, V are distinct *single variables* and S is a subset of the other two variables (possibly empty). Exclude trivial relations and do not count symmetric duplicates. (9 marks)